

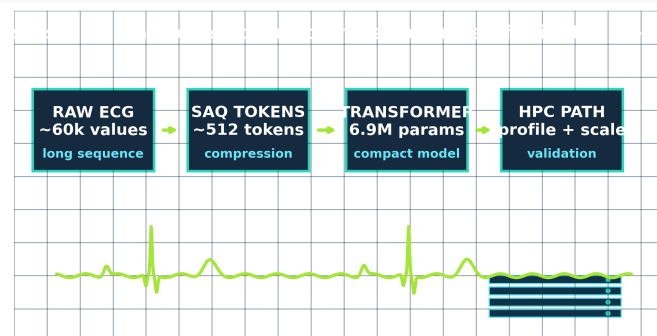
Compute-Efficient Tokenization for Long-Sequence Human Biosignal AI

Segment-Aware Quantisation for compact, calibrated ECG transformers

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Main message

ECG is used as a demanding long-sequence AI testbed. A raw 12-lead record contains tens of thousands of values. Segment-Aware Quantisation converts cardiac morphology into compact tokens, enabling a small calibrated transformer to preserve performance while reducing sequence burden.



System view: raw biosignal sequence -> compact tokens -> lightweight model -> scalable validation workflow.

Research question: can physiology-aware tokenization make long biosignal AI smaller, faster and still reliable?

~60k

raw values
12-lead, 10 sec

~512

SAQ tokens
fixed sequence

6.9M

parameters
compact model

47 ms

T4 latency
end-to-end

0.959

AUROC
PTB-XL test

0.851

CPSC AUROC
frozen transfer

Retrospective experimental findings only. Clinical evaluation is required before patient-care use.

Problem: Long Biosignals Create Compute Bottlenecks

Sequence length

Raw ECG waveforms are long structured signals. Direct modelling increases memory, latency and validation cost.

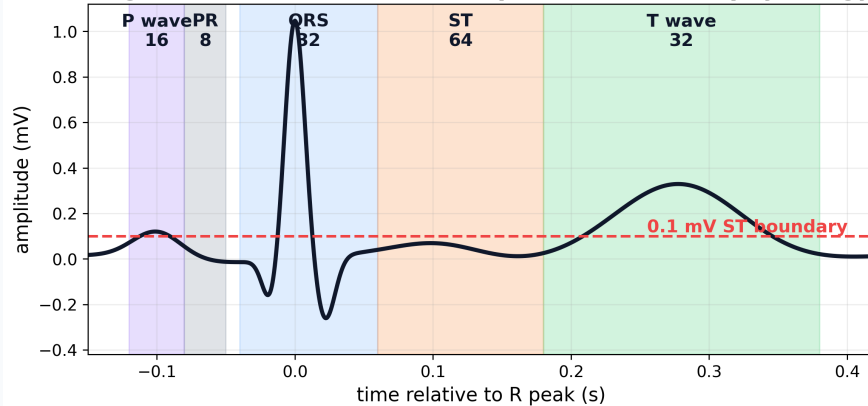
Representation fidelity

Compression must preserve tiny diagnostic morphology, especially ST-segment changes around 0.1 mV.

System reliability

A deployable model needs calibration, threshold control, robustness and latency reporting - not only accuracy.

Segment-Aware Quantisation: precision follows physiology



SAQ allocates more precision to morphology-critical segments, then downstrides to a compact token sequence.

58-75%

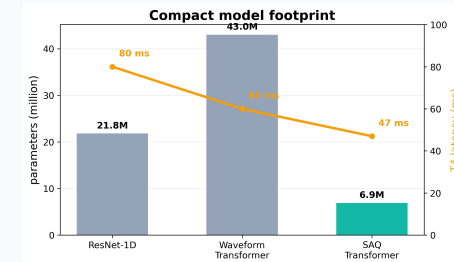
sequence reduction
reported range

0.082 mV

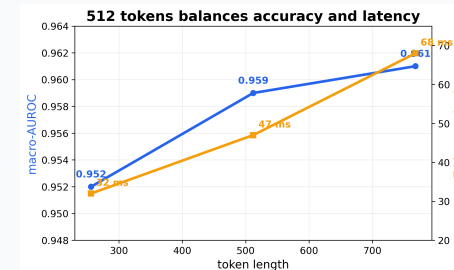
ST RMSE
below 0.1 mV

380 MB

runtime memory
reported



Compute footprint: fewer parameters and lower latency than controlled internal baselines.



Token-length trade-off: 512 tokens gives a practical accuracy-latency operating point.

3 Method: SAQ + Compact Transformer



1. Representation

SAQ gives high precision to ST and QRS regions, lower precision to low-information PR regions. The model sees a structured token sequence instead of raw waveform burden.

2. Model

A six-layer transformer uses segment-specific embeddings and cycle-aware positions. The design keeps the model compact: approximately 6.9M parameters.

3. Calibration

Temperature scaling reduces expected calibration error from 0.433 to 0.127. Thresholds are selected on validation data, including a STEMI sensitivity floor.

Audience takeaway

This is a representation-learning system for long physiological sequences: compress intelligently, keep morphology, calibrate probabilities, then measure efficiency and robustness.

Evidence: Accuracy, Calibration and Robustness

0.959

Macro-AUROC
PTB-XL test

0.921

Macro-AUPRC
PTB-XL test

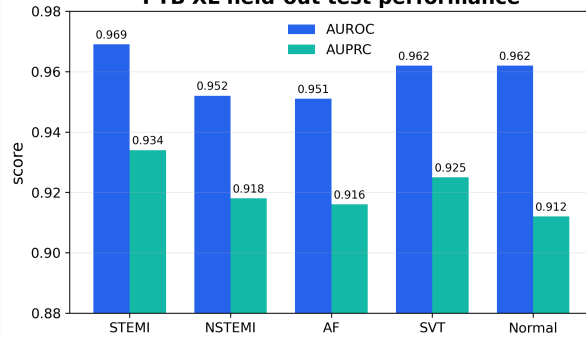
95.2%

STEMI sensitivity
constrained threshold

0.127

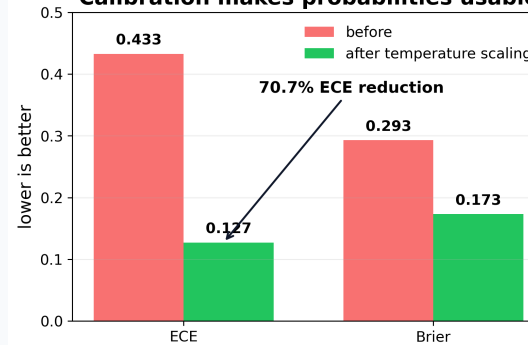
ECE after TS
calibrated output

PTB-XL held-out test performance



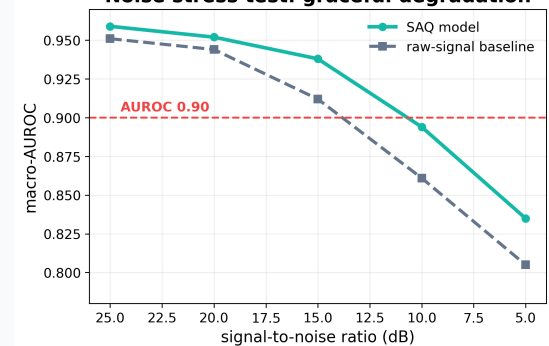
Performance: high discrimination across five emergency-relevant ECG categories.

Calibration makes probabilities usable



Calibration: probability quality improves without changing discrimination.

Noise stress test: graceful degradation



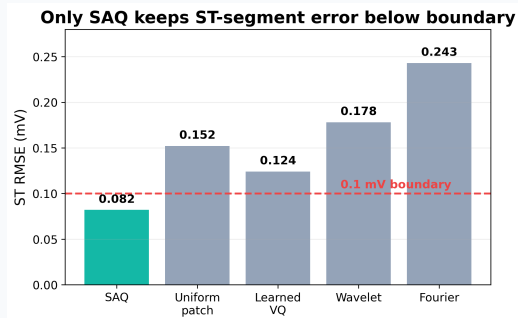
Robustness: SAQ degrades more gracefully under noise than a raw-signal baseline.

Evidence stack

The result is not just an AUROC number. The system reports compactness, calibration, threshold behaviour, transfer testing and noise degradation.

Research boundary

These are retrospective experiments. The claim is computational: efficient and reliable modelling of long biomedical sequences. It is not a clinical deployment claim.



Compression fidelity: SAQ is the only matched method below the 0.1 mV ST-segment boundary.

Why this belongs at IPCSS

- Long-sequence AI: ECG modelling is sequence compression and representation learning.
- Performance engineering: token length, memory footprint and latency affect feasibility.
- Scalable validation: larger cohorts require reproducible workflows and controlled sweeps.
- HPC next step: profile, package and scale experiments across larger compute environments.

Next technical steps

GPU profiling

Measure latency, memory and batch-size scaling across available nodes.

Distributed HPO

Run controlled ablation and hyperparameter sweeps with reproducible manifests.

Workflow packaging

Containerise preprocessing, training, calibration and evaluation.

External validation

Scale testing across datasets, acquisition devices and patient subgroups.

Limitations: retrospective design; custom taxonomy; fixed windows; no clinical deployment claim; HPC scaling is future work.